

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions
Assessment Tool Weightage: 40%
SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

AIM

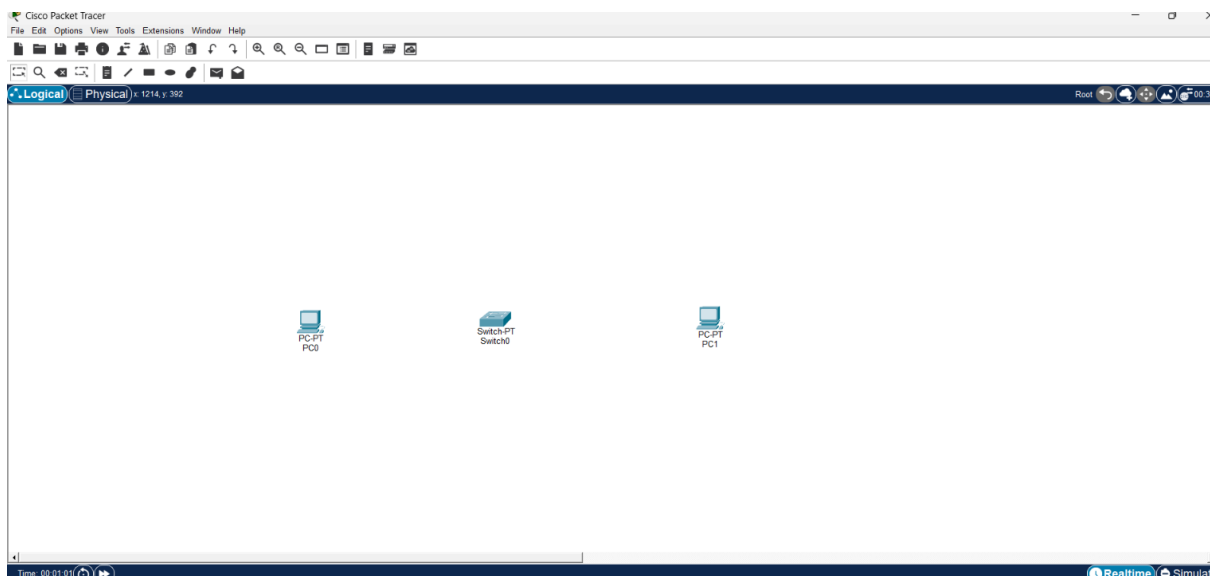
To design and configure a simple Local Area Network (LAN) consisting of two PCs connected through a switch using Cisco Packet Tracer, assign appropriate IPv4 addresses to the hosts, verify connectivity using the ping command, and analyze the Ethernet frame using Wireshark by identifying the Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). The experiment also demonstrates how the Data Link Layer uses MAC addressing for successful frame delivery within a local network.

PROCEDURE

1. Open **Cisco Packet Tracer** and place two PCs (**PC0** and **PC1**) and one **Cisco 2960 switch** in the workspace.
2. Connect **PC0** and **PC1** to the switch using **Copper Straight-Through** cables.

3. Configure **PC0** with IP address 192.168.1.1 and subnet mask 255.255.255.0, and configure **PC1** with IP address 192.168.1.2 and subnet mask 255.255.255.0.
4. Open the Command Prompt of PC0 and execute ping 192.168.1.2 to verify connectivity between the two PCs.
5. Open **Wireshark**, select the active network interface, start packet capture, and generate ping traffic for analysis.
6. Apply the icmp display filter in Wireshark and examine the captured Ethernet frame to identify the **Source MAC Address**, **Destination MAC Address**, **EtherType**, and **FCS** fields.
7. Analyze how the **Data Link Layer** uses MAC addresses and the switch's MAC address table to forward Ethernet frames to the correct device within the local network.

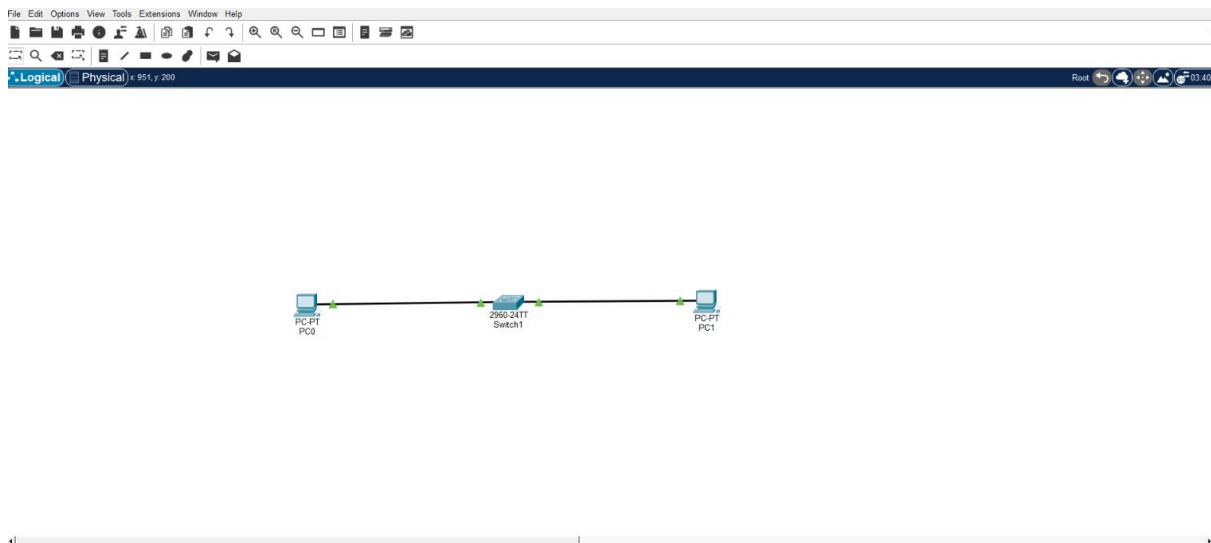
STEP 1: Design a simple LAN consisting of two PCs



EXPLANATION:

1. Open Cisco Packet Tracer.
2. Select End Devices and add two PCs.
3. The PCs will appear as PC0 and PC1.
4. Place the two PCs in the workspace to create the basic LAN setup

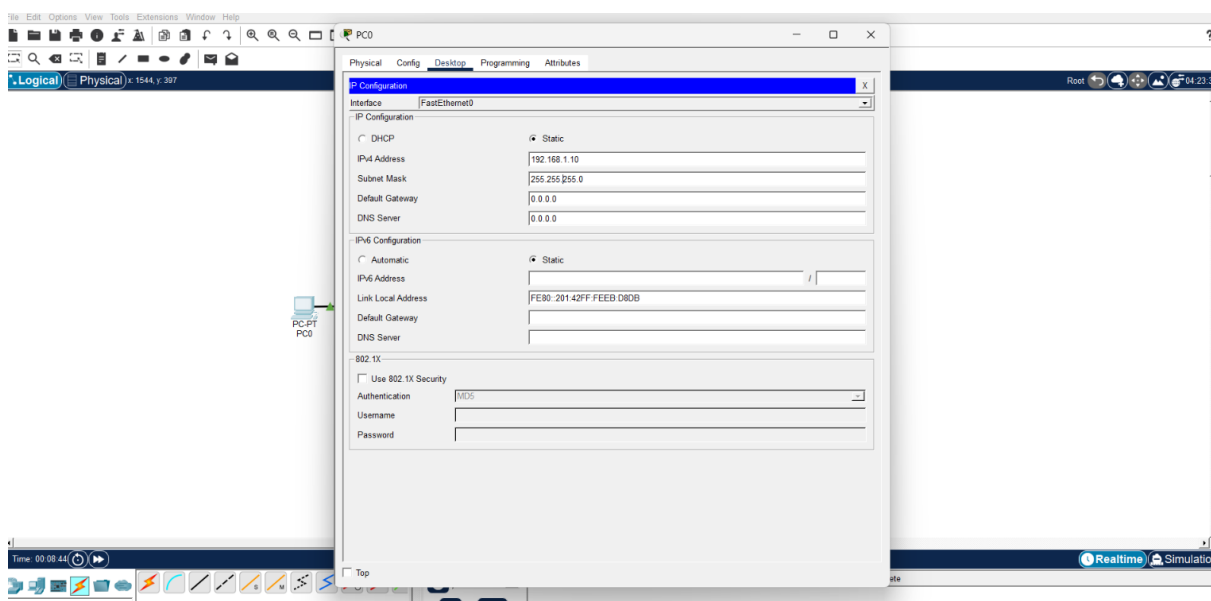
STEP 2: Connect the PCs Through a Switch



EXPLANATION:

1. In Cisco Packet Tracer, select Network Devices → Switches.
2. Add a 2960 switch between PC0 and PC1.
3. Select Copper Straight-Through cable and connect PC0 → Switch and PC1 → Switch.
4. Wait until the connection lights become green, confirming the PCs are connected through the switch.

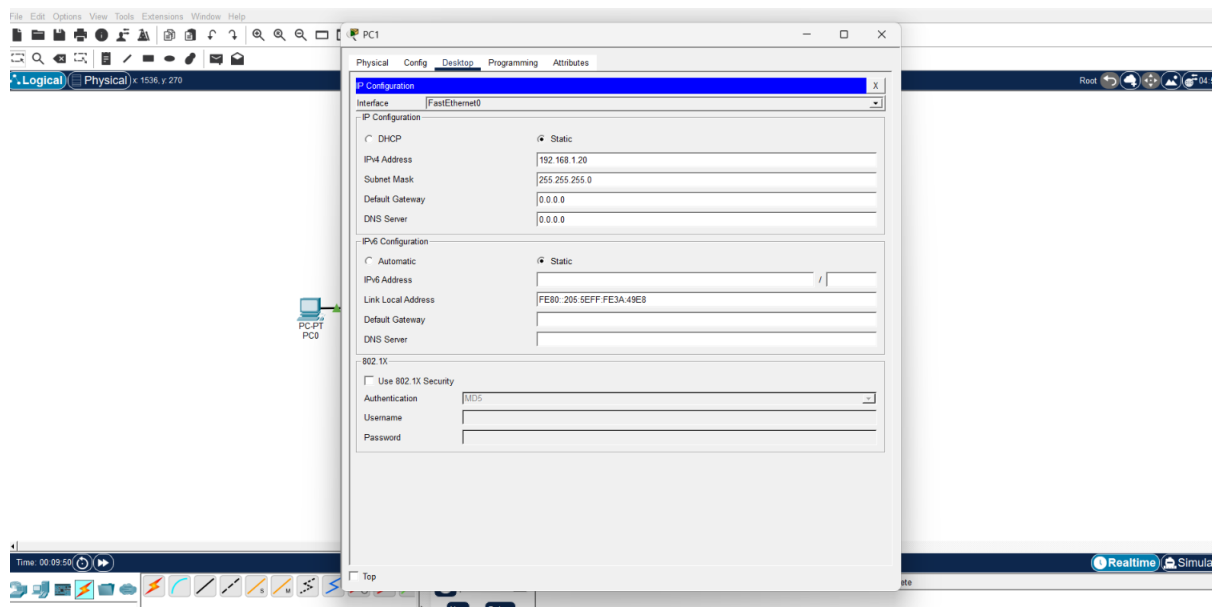
STEP 3: PC0 IP Configuration



EXPLANATION:

- Open PC0 in Cisco Packet Tracer and select Desktop.
- Click IP Configuration and select Static.
- Enter IP Address as 192.168.1.10.
- Enter Subnet Mask as 255.255.255.0 and close the window.

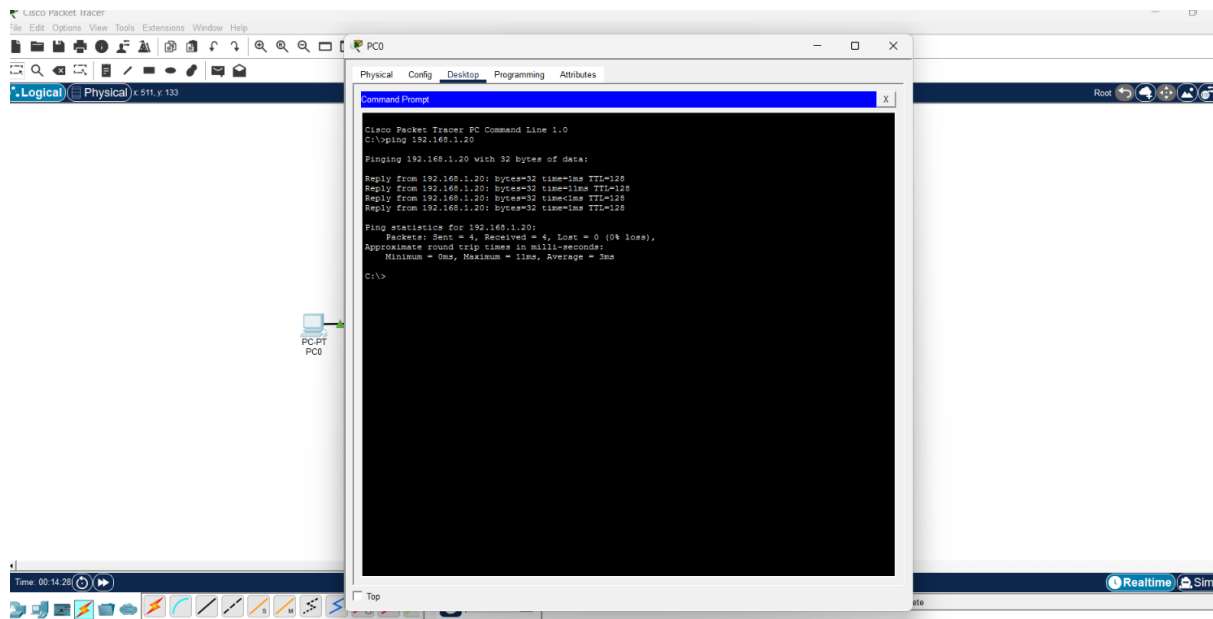
STEP 4: PC1 IP Configuration



EXPLANATION:

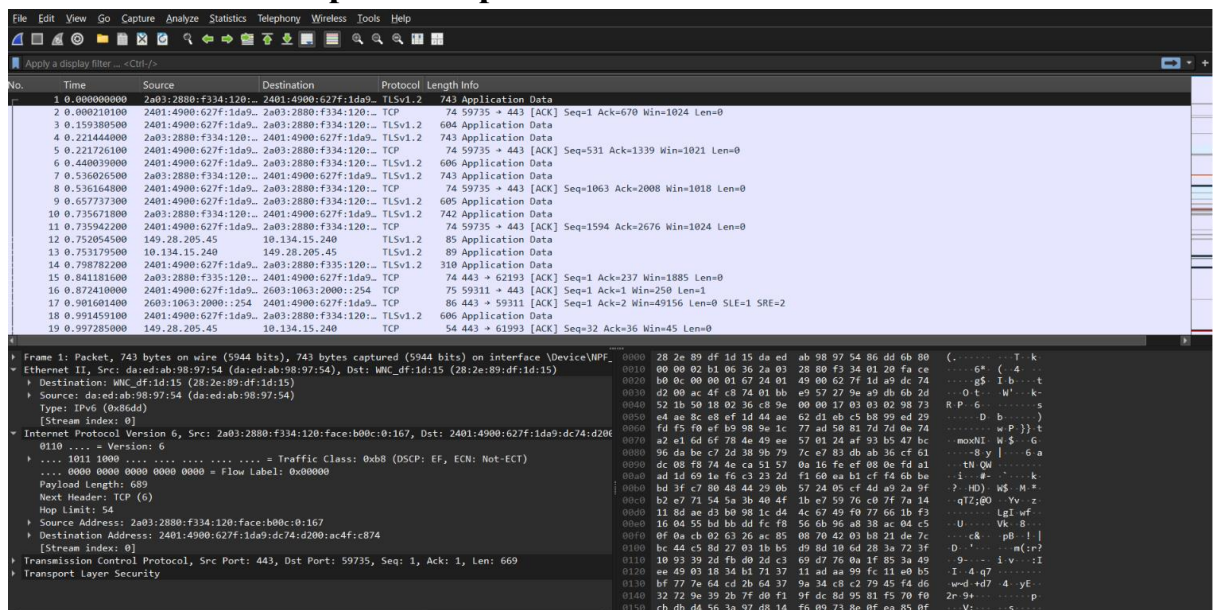
- Open PC1 in Cisco Packet Tracer and select Desktop.
- Click IP Configuration and select Static.
- Enter IP Address as 192.168.1.20.
- Enter Subnet Mask as 255.255.255.0 and close the window.

STEP 5: Ping/Command Prompt screenshot



EXPLANATION:

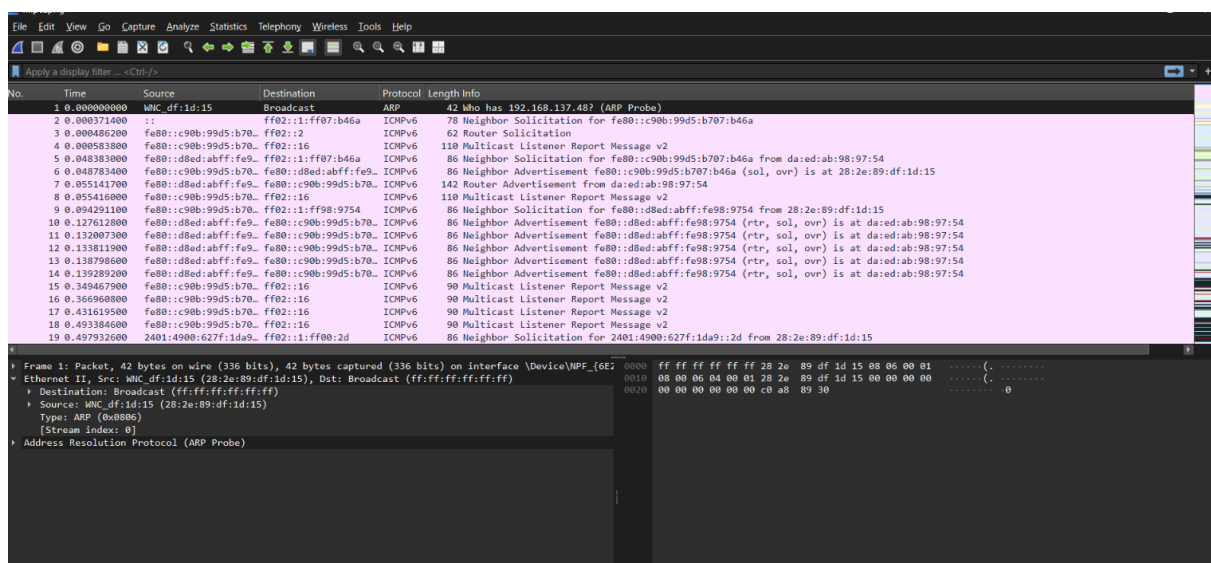
- Open PC0 in Cisco Packet Tracer and go to Desktop → Command Prompt.
- Type the command ping 192.168.1.20 and press Enter.
- Check that you receive Reply from 192.168.1.20 messages.
- Verify the result shows Sent = 4, Received = 4, Lost = 0 (0% loss)
- **STEP 6: Wireshark packet capture**



EXPLANATION:

- Open Wireshark and start packet capturing on the active network interface.
- Network packets are displayed in the packet list with Source, Destination, Protocol, and Length.
- Select a packet to view its detailed information in the Packet Details section.
- The Ethernet II section shows the Source MAC Address, Destination MAC Address, and Type/EtherType.
- The captured packets can be analyzed to understand how data is transmitted between devices in the network.

STEP 7: Ethernet frame analysis



EXPLANATION:

- Select an Ethernet packet in Wireshark.
- Open the Ethernet II section.
- Check the Source MAC Address of the sender.
- Check the Destination MAC Address of the receiver.
- Check the Type/EtherType to identify the network protocol.

STEP 8: The Data Link Layer/MAC explanation

- The Data Link layer is responsible for delivering data between devices in the same local network.
- Each network device has a unique MAC address that identifies the device.
- The Source MAC address identifies the device that sends the Ethernet frame.
- The Destination MAC address identifies the device that should receive the frame.
- The switch checks the destination MAC address and forwards the frame to the correct port, ensuring successful delivery within the LAN.

Simple Example:

PC0 ————— Switch ————— PC1

Sender

Receiver

Source MAC

Destination MAC

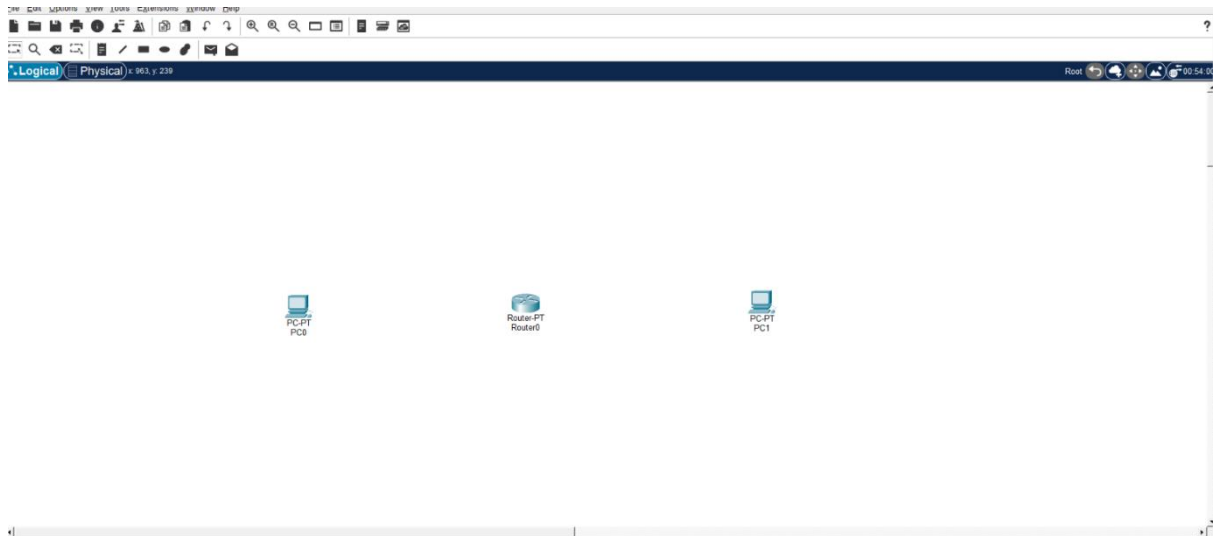
Aim:

To develop a simple network consisting of two PCs connected through a router using Cisco Packet Tracer, configure the required IP addresses, verify connectivity using the ping command, capture ICMP traffic using Wireshark, and analyze Echo Request, Echo Reply, ICMP fields, and Round Trip Time (RTT).

Procedore:

1. Design a simple network consisting of two PCs connected through a router using Cisco Packet Tracer.
2. Connect PC0 and PC1 to the router using suitable Ethernet cables.
3. Configure the IP address, subnet mask, and default gateway for PC0 and PC1.
4. Configure the router interfaces with suitable IP addresses and enable the interfaces.
5. Verify communication between PC0 and PC1 using the ping command.
6. Capture the ICMP packets using Wireshark and identify the Echo Request and Echo Reply messages.
7. Analyze the ICMP fields such as Type, Code, Identifier, Sequence Number, and calculate the Round

STEP1: Develop a simple network consisting of two PCs



EXPLANATION:

- Open Cisco Packet Tracer.
- Select End Devices and add two PCs.
- Select Network Devices → Routers and add one router.
- Place them as PC0 → Router → PC1.

STEP2: Connect the Devices



EXPLANATION:

1. Open Cisco Packet Tracer.
2. Add two PCs, PC0 and PC1.
3. Add one 2911 router.
4. Connect both PCs to the router.
5. Check that all connections are active.

Step 2: Configure PC0 IP Address



EXPLANATION:

- Open PC0 → Desktop → IP Configuration.
- Select the Static option.
- Enter IP address 192.168.1.1.
- Enter subnet mask 255.255.255.0.
- Enter default gateway 192.168.1.254.

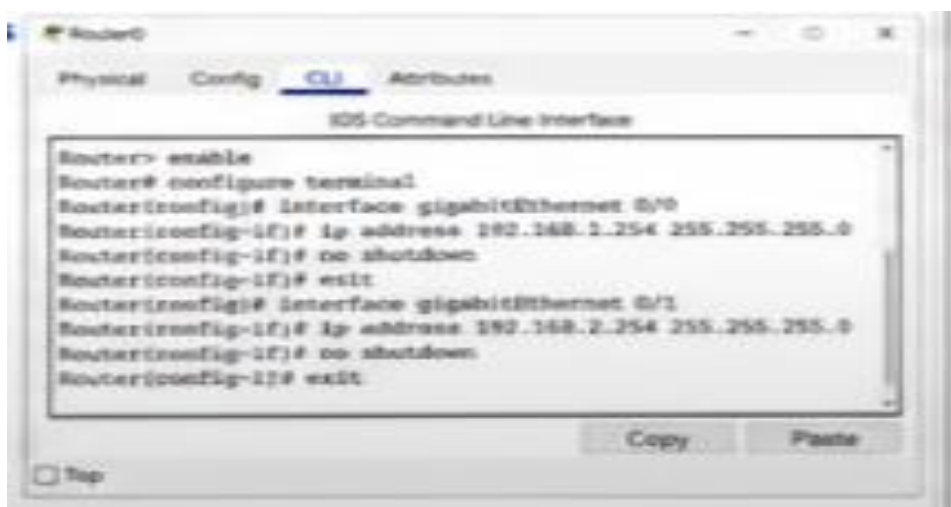
Step 3: Configure PC1 IP Address



EXPLANATION:

1. Open PC1 → Desktop → IP Configuration.
2. Select the Static option.
3. Enter IP address 192.168.2.1.
4. Enter subnet mask 255.255.255.0.
5. Enter default gateway 192.168.2.254.

Step 4: Configure Router Interfaces



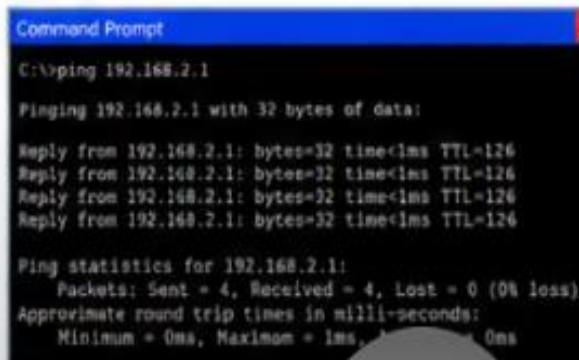
EXPLANATION:

1. Open the router and select the CLI tab.
2. Configure the first interface with 192.168.1.254.
3. Configure the second interface with 192.168.2.254.
4. Set the subnet mask to 255.255.255.0.
5. Use no shutdown to activate both interfaces.

Step 5: Test PC0 to Router Connectivity

STEP 10: Test Connectivity (Ping)

- Click PC0 > Desktop > Command Prompt
- Ping PC1 IP: 192.168.2.1
- You should get replies.



```
Command Prompt
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

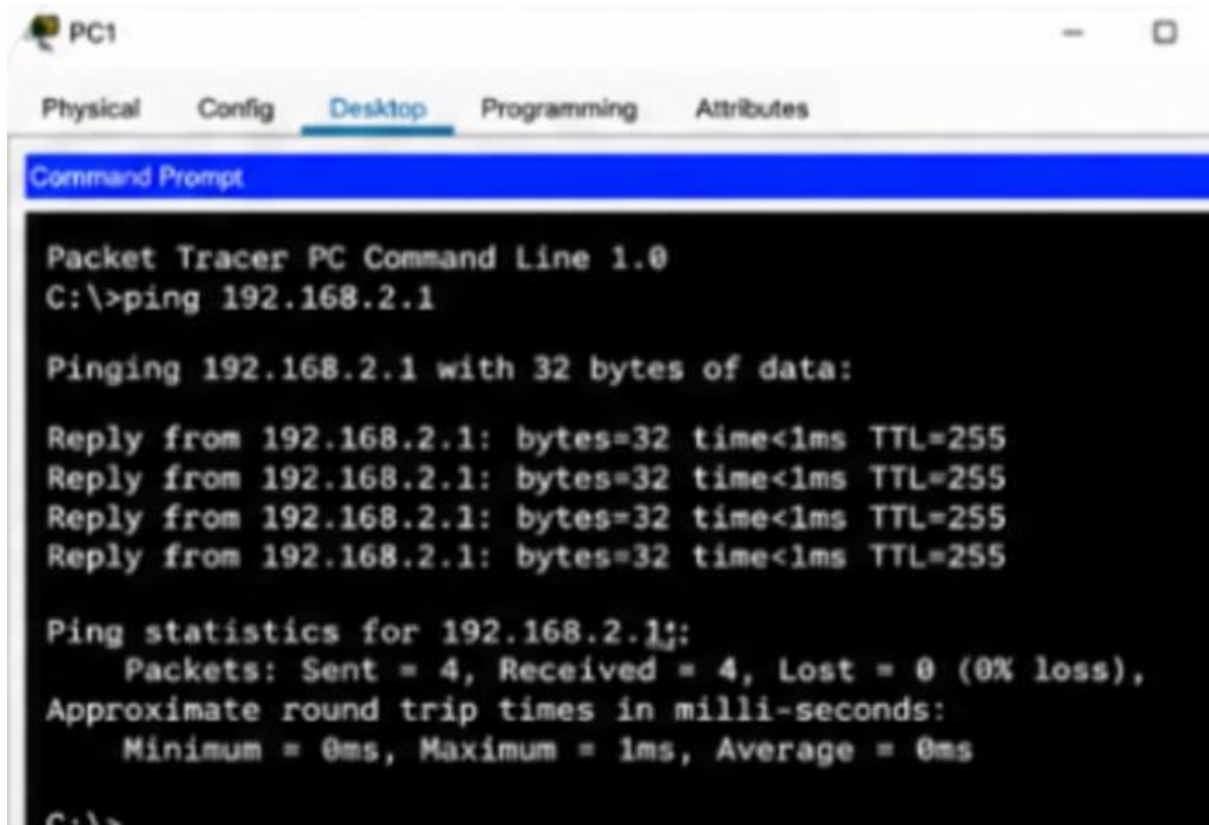
Reply from 192.168.2.1: bytes=32 time<1ms TTL=126
Reply from 192.168.2.1: bytes=32 time<1ms TTL=126
Reply from 192.168.2.1: bytes=32 time<1ms TTL=126
Reply from 192.168.2.1: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

EXPLANATION:

1. Open PC0 → Desktop → Command Prompt.
2. Use the ping command.
3. Ping the router address 192.168.1.254.
4. Check whether reply messages are received.
5. Successful replies show that PC0 can reach the router.

Step 6: Test PC1 to Router Connectivity



The screenshot shows the Packet Tracer interface for PC1. The 'Desktop' tab is selected, displaying a Command Prompt window. The command prompt shows the execution of the 'ping 192.168.2.1' command, which successfully pings the destination IP address. The output includes four successful replies and a summary of the ping statistics.

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

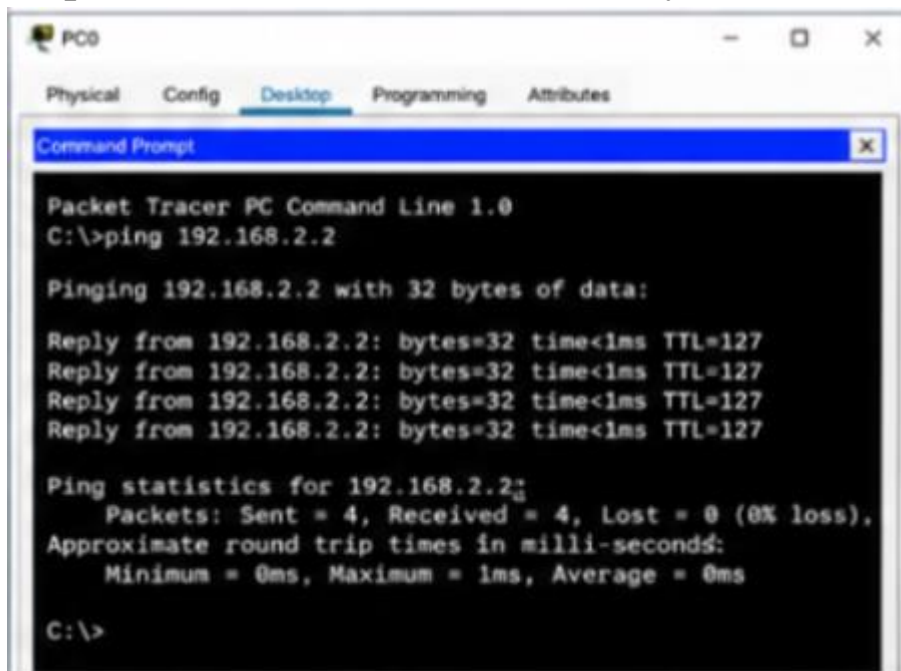
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
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Reply from 192.168.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

EXPLANATION:

1. Open PC1 → Desktop → Command Prompt.
2. Use the ping command.
3. Ping the router address 192.168.2.254.
4. Check whether reply messages are received.
5. Successful replies show that PC1 can reach the router.

Step 7: Test PC0 to PC1 Connectivity



EXPLANATION:

1. Open the command prompt on PC0.
2. Enter ping 192.168.2.1.
3. The ping request travels through the router.
4. Check for successful reply messages.
5. Successful replies confirm PC0-to-PC1 connectivity.

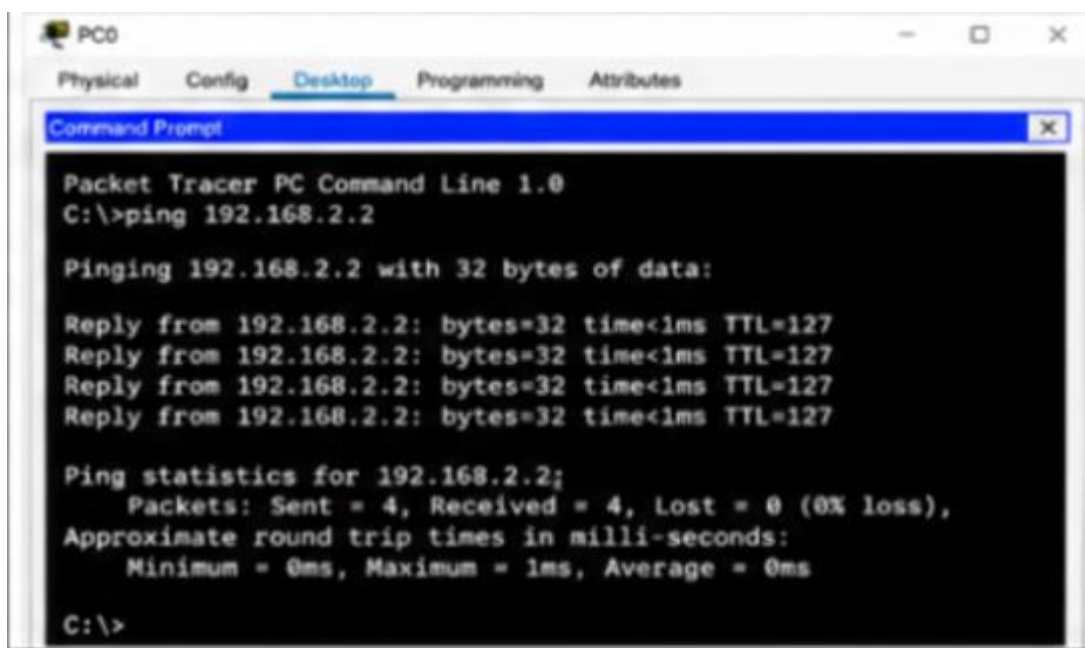
Step 8: Enable Simulation Mode and ICMP Filter



EXPLANATION:

1. Select Simulation Mode in Packet Tracer.
2. Open the Event List Filters.
3. Select the ICMP protocol.
4. Remove unnecessary protocol filters if required.
5. ICMP packets will now be displayed.

Step 9: Generate ICMP Packet



The screenshot shows a Packet Tracer PC0 window with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the output of a ping command. The text in the command prompt is as follows:

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

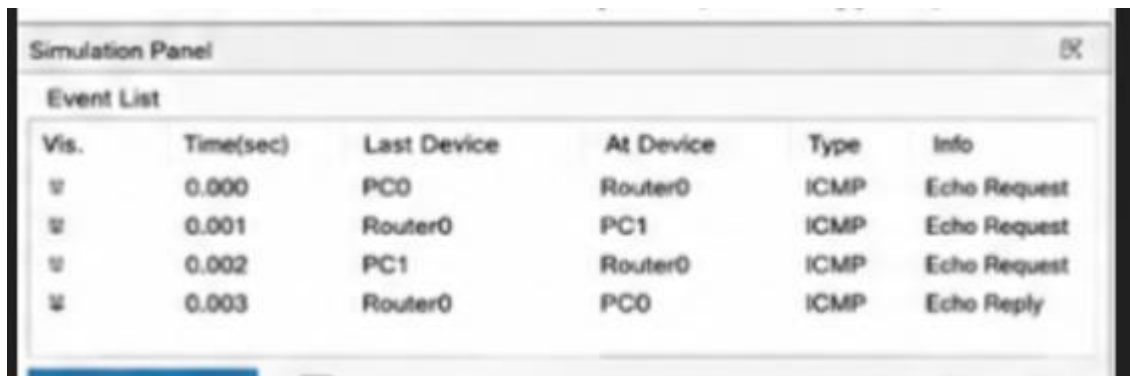
Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

EXPLANATION:

1. Open the command prompt on PC0.
2. Enter ping 192.168.2.1.
3. The ping command generates ICMP packets.
4. An Echo Request is sent from PC0.
5. PC1 sends an Echo Reply back to PC0.

Step 10: Observe ICMP Echo Request



The image shows a 'Simulation Panel' window with a title bar and a close button. Inside, there is an 'Event List' table. The table has six columns: 'Vis.', 'Time(sec)', 'Last Device', 'At Device', 'Type', and 'Info'. It contains four rows of data representing ICMP Echo Request and Reply packets between PC0, Router0, and PC1.

Vis.	Time(sec)	Last Device	At Device	Type	Info
W	0.000	PC0	Router0	ICMP	Echo Request
W	0.001	Router0	PC1	ICMP	Echo Request
W	0.002	PC1	Router0	ICMP	Echo Request
W	0.003	Router0	PC0	ICMP	Echo Reply

EXPLANATION:

1. Observe the packet travelling from PC0 to PC1.
2. The packet is identified as ICMP Echo Request.
3. The Echo Request has Type 8.
4. The Code value is 0.
5. This packet checks whether PC1 is reachable.

Step 11: Observe ICMP Echo Reply



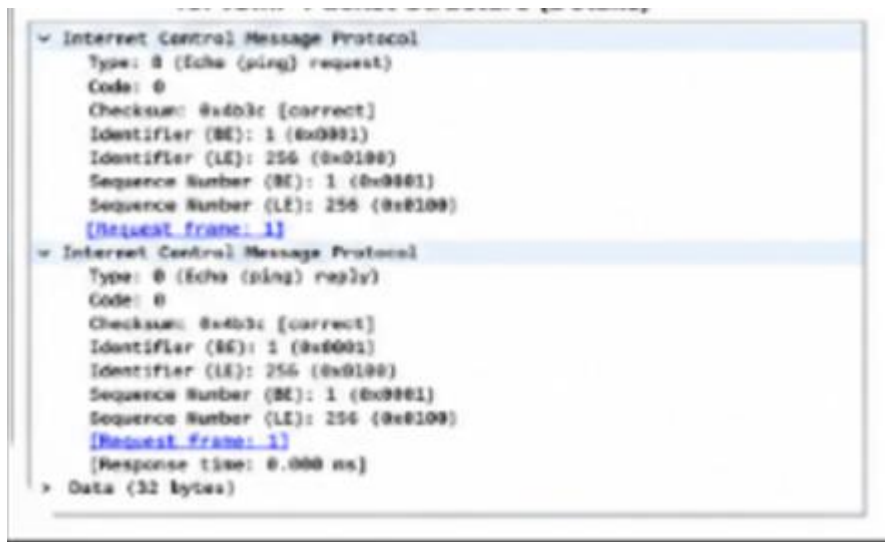
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Vis.	Time(sec)	Last Device	At Device	Type	Info
W	0.000	PC0	Router0	ICMP	Echo Request
W	0.001	Router0	PC1	ICMP	Echo Request
W	0.002	PC1	Router0	ICMP	Echo Reply
W	0.003	Router0	PC0	ICMP	Echo Reply

EXPLANATION:

1. Observe the packet travelling from PC1 to PC0.
2. The packet is identified as ICMP Echo Reply.
3. The Echo Reply has Type 0.
4. The Code value is 0.
5. The reply confirms successful communication.

Step 12: Examine ICMP Packet Fields



EXPLANATION:

1. Open the details of the ICMP packet.
2. Check the Type field.
3. Check the Code field.
4. Check the Identifier field.
5. Check the Sequence Number field.

Step 13: Calculate Round Trip Time (RTT)

14. Round Trip Time (RTT) Calculation

Packet	Event	Time (sec)
1	Echo Request (PC0 → PC1)	0.000
2	Echo Reply (PC1 → PC0)	0.002

RTT = Reply Time - Request Time
RTT = 0.002 - 0.000 = 0.002 sec = 2 ms

The screenshot shows the output of a ping command in a Windows Command Prompt. It displays the IP address being pinged (192.168.2.2), the size of the data (32 bytes), and the time taken for each of four replies (all 127 ms). It also shows the ping statistics: 4 packets sent, 4 received, 0 lost, with an average round trip time of 127 ms.

Conclusion:

The ICMP Echo Request (Type 8) is sent from PC0 to PC1.

PC1 responds with ICMP Echo Reply (Type 0).

Round Trip Time (RTT) is calculated as the difference between request and reply time.

Hence, connectivity between PC0 and PC1 is successful.

EXPLANATION:

1. Note the time of the Echo Request.
2. Note the time of the corresponding Echo Reply.
3. Subtract the request time from the reply time.
4. The result gives the Round Trip Time (RTT).
5. In the given screenshot, RTT is 0.002 sec = 2 ms.

Step 14: Result / Conclusion

- The network was successfully created in Packet Tracer.
- IP addresses were configured on both PCs and the router.
- Connectivity was successfully tested using ping.
- ICMP Echo Request and Echo Reply packets were observed.
- The RTT was calculated as 2 ms, confirming successful communication.